

Puspak Chakraborty

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EDUCATION

Indian Institute of Technology Madras (IIT Madras)

M.S. in Data Science and Artificial Intelligence

CGPA: 9.06/10

Chennai, India

June 2024 – Present

RCC Institute of Information Technology

B.Tech in Computer Science

CGPA: 9.33/10

Kolkata, India

June 2016 – June 2020

RESEARCH EXPERIENCE

Indian Institute of Technology Madras

M.S. Research Scholar

Chennai, India

June 2024 – Present

- Investigating theoretical equivalences between Deep Linearly Gated Networks and differentiable gradient boosting architectures.
- Developed Boosting architecture that uses neural routers for intelligent spatial partitioning to handle heteroscedastic noise.
- **Publications:** Puspak Chakraborty, Arun Rajkumar. “Neural Routed Boosting: Robust Learning against Heteroscedastic Noise.” *Conference on Uncertainty in Artificial Intelligence (UAI) 2026. Proceedings of Machine Learning Research (PMLR)*. [Link]

WORK EXPERIENCE

Indian Institute of Technology Madras

Teaching Assistant

Chennai, India

January 2025 – present

- **Foundations of Machine Learning (DA5400):** Facilitated course operations and provided academic support for approximately 150 students (July 2025 – Nov 2025).
- **Mathematical Foundations of Data Science (DA5000):** Assisted in course operations for about 90 students (Jan 2025 – May 2025).

Infosys

Specialist Programmer

Kolkata, India

October 2020 – July 2024

- Developed serverless data pipelines using AWS SQS-triggered Lambda functions for a legal domain product.
- Designed REST APIs and optimized existing enterprise codebase, significantly improving processing efficiency.

PROJECTS

Multi-Agent Research Assistant | *LangGraph, HuggingFace, RAG, ChromaDB* | github

- Architected a local multi-agent AI research assistant integrating retrieval from LocalPDFs, arXiv, and DuckDuckGo search tools via LangGraph, powered by a 4-bit quantized Mistral-Nemo-Instruct-2407 model for on-device inference.
- Engineered a hybrid Retrieval-Augmented Generation (RAG) pipeline integrating BM25 and dense retrieval via ChromaDB, enhanced by layout-aware chunking with Marker.
- Implemented robust LLMops tracking and observability utilizing Arize Phoenix for model evaluation and trace monitoring.

Polyvox: Optimized Edge AI Acoustic Pipeline | *PyTorch, Pyannote, torchaudio, Knowledge Distillation* | github

- Engineered a highly optimized edge AI acoustic processing pipeline utilizing a dynamic routing system and Pyannote diarization.
- Trained and finetuned Sandglasset model on customized dynamic LibriMix dataset.
- Enhanced the intent classification engine's robustness by generating custom datasets incorporating comprehensive slots and phonetic error dictionaries.

Handdrawn Image Classifier System with Feedback | *FastAPI, MLflow, Docker, Prometheus, DVC* | github

- Architected a containerized FastAPI web app for real-time doodle classification with human-in-the-loop feedback.
- Built an automated retraining pipeline using DVC, Git, and MLflow for seamless dataset versioning and model updates.
- Implemented system observability with Prometheus, Grafana, and a PostgreSQL-backed dashboard for pipeline tracking.

Seq2Seq Neural Machine Transliteration | *PyTorch, WandB, RNN/LSTM/GRU, Attention* | github | WandB

- Developed an Encoder-Decoder model (RNN/GRU/LSTM with Attention) for English to Hindi transliteration.
- Implemented greedy and beam search decoding strategies with length penalties.
- Orchestrated Bayesian hyperparameter tuning via Weights & Biases and built custom attention heatmap visualizations.

SKILLS

Languages: C++, Python, SQL

Tools & Frameworks: PyTorch, LangGraph, HuggingFace, AWS, Git, MySQL, Prometheus, Grafana, Airflow, DVC, ChromaDB

Core Concepts: Machine Learning, Artificial Intelligence, Optimization Algorithms, LLMops, RAG

ACHIEVEMENTS & CERTIFICATIONS

- Secured Rank 11 out of 61 teams in KLA Image Restoration Hackathon, hosted by Computer Science Department, IIT Madras.
- Secured **All India Rank (AIR) 439**/123,967 candidates in **GATE 2024** (Computer Science), **GATE Score:** 765/1000.
- AWS Certified Cloud Practitioner (01/2022 – 01/2025).